

Review

A review of image processing and quantification analysis for solid oxide fuel cell

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HIGHLIGHTS

- Detail methodologies of image processing and quantification methods of SOFC are discussed.
- Pros and cons of image processing and quantification methods for SOFC microstructural analysis are highlighted.
- Quantification accuracy of microstructural properties are validated on SOFC.

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ABSTRACT

The purpose of this study is to investigate the approaches applied to analyze solid oxide fuel cell (SOFC) microstructural properties. Both manual and automated image processing approaches applied on SOFC microstructural images which are obtained from several types of tomography such as dual-beam focused ion beam with scanning electron microscopy (FIB-SEM), Electron Backscatter Diffraction (EBSD) and others are discussed. In fact, to achieve a realistic and accurate SOFC microstructural properties, such as average diameter, volume fraction, triple phase boundary (TPB), area interface density and tortuosity factor, the approaches of image processing and quantification are crucial for a reliable image generation for quantification purposes. The microstructural properties are optimized to improve SOFC electrode performance. Therefore, the image processing and quantification approaches are outlined and reviewed. Despite the automated image processing and quantification algorithms significantly outperform manual image processing and quantification approaches in terms of computing speed when evaluating and measuring microstructural properties, the efficiency and productivity are still extremely taken into concern. As a result, image processing and quantification approaches are concluded and presented respectively in this paper.

1. Introduction

SOFC is widely employed to generate electricity efficiently from fuels, for example, methane, hydrogen, ammonia and natural gas. Due to SOFC has high adaptability and excellent energy conversion efficiency, In a high temperature SOFC that is operated above 800 °C, the most popular materials used for interconnection are high temperature metal or doped lanthanum chromite, yttria-stabilized zirconia (YSZ) for electrolyte, strontium-doped lanthanum manganite (LSM) for cathode, and nickel (Ni)/YSZ for anode [1–3]. However, the selection of materials are strongly influenced by the operation temperature of SOFC. Depending on particular cell configurations and system architecture,

SOFC operates at the temperature, starting from 700 °C to 1000 °C under pressurized conditions or atmospheric pressure [1]. On the other hand, high molecular weight hydrocarbon fuels such as methane and natural gas can be directly supplied to SOFC in order to facilitate rapid electro catalysis and oxide diffusion through electrolyte [4]. As a result, there is no longer a need for an expensive external reformer system [5]. Although SOFC works well in static or portable heat and power generation, it also confronts material challenges in terms of developing feasible and long-lasting commercial or military device application due to sustainability of structural and chemical integrity of materials are highly concerned in SOFC design [6]. On the contrary, SOFC can utilize

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a wide variety of hydrocarbon fuels in conjunction with a conventional power generation system without any noise disturbance. As mentioned earlier, SOFC must work under high temperature in pressurized condition to replace the combustor in steam or gas turbine. As such, the corresponding SOFC combined with turbines yields an efficiency over 70% [4].

Nowadays, numerous types of imaging methods that are heavily influenced by material properties are employed for microscopic imaging, for example, scanning electron microscopy (SEM) imaging, backscattered electron (BSE) imaging, optical microscopy (OM) imaging, and others. These imaging methods allow for the direct observations of numerous sequential two-dimensional (2D) images of porous material and three-dimensional (3D) computational reconstruction structure as well as evaluating microstructural properties from the 3D reconstructed structure without making any assumptions based on 2D images [7]. Apparently, image quality in terms of contrast, resolution and size are highly concerned especially in image processing and quantification [8]. In order to obtain high quality images, imaging circumstances such as working imaging distance, operating voltage and current also needed to be taken care [8]. Furthermore, Wilson et al. [9] suggested using FIB-SEM to analyze the porosity of SOFC. With the aid of such tomography, a series of consecutive 2D SEM images were yielded and stacked to reconstruct into 3D microstructural volume for the ease of volumetric information quantification. Next, Gunda et al. [10] reconstructed porous and ceramic LSM cathode electrode material for SOFC through dual-beam FIB-SEM tomography. After acquiring 2D microstructural images by FIB-SEM tomography, the images were processed using a formulated methodological approach for 3D reconstruction. Then, using normalized effective transport properties in x-, y-, and z-directions, microstructural analysis was conducted on computational domains that were reconstructed by FIB-SEM tomography to calculate area of internal surface and porosity [10]. The benefit of FIB-SEM tomography is it can only be acquired at low magnifications, allowing the element from the materials to be revealed even when it is surrounded by several undesirable substances such as void structures and tiny pores [10]. Based on scanning electron micrographs analysis by Hwang et al. [11], two distinct artifacts, namely curtain and edge effects, manifest as a result of charging from high-energy electron beam illumination. The presence of line patterns stemming from a non-uniform FIB milling process and edge effects arising due to electrical charging encountered by scanning electron micrographs [11]. Curtain effect occurs when the ion beam deviates from perpendicular incidence on the sample during FIB-SEM tomography. As the ion beam scans across the specimen, non-uniform material removal results in thickness variations on the microstructural images. To mitigate the curtain effect, Prokop et al. [12] implemented a preventive measure by applying a thin platinum film. This film was deposited onto the catalytic layer through the injection of a platinum precursor from the gas injection system [12]. To address potential misalignment arising from the beam or sample shifts, markers were strategically inscribed on the catalytic layer and positioned outside the platinum layer. This strategic placement facilitated the autofocus of lenses and streamlined other automated correction procedures, ensuring maintenance of optimal imaging conditions. Furthermore, a concise acquisition period was adopted for each image. This deliberate choice not only curtailed the impact of image drift but also resulted in an enhancement of overall image fidelity [12]. In the realm of FIB-SEM tomography, the intriguing phenomenon of shadowing arises not merely from image drift but also from inadequate electron signal transmission to the detectors [13]. A notable illustration of this occurs with steep edges possess the capability to cast shadow that disrupt the trajectory of the ion beam, leads to the appearance of darker regions within the images [13]. The shadowing effect becomes pronounced when edges of surrounding material obstruct the cross-sections, effectively blocking the electron beam's path [14]. To confront the intricacies of shadowing, a precisely carved U-shaped trench encircles the cross-section, skillfully milled to diminish any lingering shadows. The process

culminates with a meticulous polishing of the cross-section, ensuring a flawlessly refined result [14]. Besides that, shearing emerges as a prevalent challenge in FIB-SEM tomography. As highlighted in Zhang's research [15], shearing deformation unfolded when the resin material experienced micro-distortion due to the repetitive ion bombardment. The resultant shearing deformation, observable in both YZ and XZ perspectives, originated from the micro-distortion of the resin during the slice-and-view procedure, resulted 3D reconstruction model to introduce substantial errors into the derived surface statistics [15].

Liu and Jiao [16] advocated using FIB/lift-out tomography to analyze microstructural degradation of the anode or electrolyte interface in SOFC. Transmission electron microscopy (TEM) was used to prepare the SOFC anode or electrolyte interface in prior to conduct the experiment. In the realm of interfacial microstructures nanoscale investigations, TEM is the favored analytical method, but it is severely constrained by the challenges of specimen preparation especially in interface study of SOFC. Yet, FIB overcomes the restriction by enabling submicron precision TEM specimen creation from a specific region as well as uniform homogeneous thinning of structurally and chemically distinct heterogeneous microstructures [16]. Furthermore, according to Thyden et al. [17], SEM tomography provided microstructural images of SOFC Ni-YSZ anode with relatively similar backscatter coefficients at normal acceleration voltages (10–20 kV). As a result, a lack of distinction between phases in images was happened. Therefore, by using image analysis tools, it is impossible to evaluate and estimate the area fraction or particle size of phases [17]. They also mentioned that OM can boost the contrast, but at the same time affects the image resolution. Not only that, X-ray mapping is suitable to define the interface between Ni and YSZ, but it necessitates a significant amount of time and has a poor spatial and depth resolution [17]. To overcome the difficulties mentioned above, Thyden et al. [17] proposed a low voltage field emission FE-SEM that provides sufficient image intensity and resolution to achieve desired contrast between Ni and YSZ. Additionally, Monachon et al. [18] acquired images of BSE SOFC Ni-YSZ cermet using high voltage BSE-SEM tomography. In order to prevent problems caused by grinding residues, the specimen was polished without impregnation prior to high voltage BSE tomography [18]. As a conclusion, the difference in backscattering coefficients between Ni and YSZ is increased while also improving contrast. Nonetheless, the disadvantages of applying such procedures are low resolution and high penetration of electrons in the material that will affect the volume fraction of Ni and YSZ [18]. Besides that, the identification of distinct phases in BSE microstructural image becomes feasible when variations exist in the average atomic number of the components such as Ni, Pore and YSZ. However, challenges arise when achieving adequate contrast between phases is hindered by closely aligned average atomic number. This poses a segmentation challenge, particularly in scenarios where the average atomic number of the phases such as LSM and YSZ are in close proximity. To address this issue, ongoing research efforts are dedicated to the development of diverse mathematical methodologies for brightness histograms analysis and image segmentation. These approaches aim to provide quantitative insights into pore sizes and phase characteristics of LSM, YSZ and others.

On the other hand, Wang et al. [19] employed EBSD to assess the cross-sectional distribution of grains within the insulation layers following thermal cycles. Based on Ko et al. [20] analysis of EBSD, they emphasized the efficacy of EBSD as a formidable tool for elucidating the crystallographic phase and microstructural intricacies in materials. Despite its commendable capabilities, the seamless adoption of EBSD confronts hurdles when analyzing small-grained oxide layers [20]. Besides that, Shen et al. [21] proposed a pioneering deep learning approach by leveraging the sophisticated U-net architecture and centered around EBSD. This approach establishes a meticulously crafted and cost-effective framework for classifying and quantifying SEM images. Specifically designed for SEM images, this method seamlessly

accommodates a diverse range of image qualities (imaging rate: 100–1000 ns/pixel) and magnifications (500–4000x) [21]. This versatile methodology adeptly captures complex features, facilitating precise pixelwise segmentation of SEM images. Furthermore, it unfolds into three primary applications within the domain of deep learning. Firstly, it excels at categorizing complex phases in SEM images, presenting an efficient alternative that bypasses the need for extensive experimental efforts often associated with quasi-EBSD map production. Secondly, the method expedites the generation of EBSD phase maps by seamlessly integrating the model with the OpenCV package [22], enabling rapid microstructure quantification. This innovative methodology propels EBSD analysis by swiftly producing EBSD phase maps from regular SEM images, eliminating the necessity for time-consuming experimental procedures [21]. Besides that, Heenan et al. [23] proposed a non-destructive three phase segmentation method for Ni-YSZ electrodes using lab-based X-ray nano-CT. The wider availability of lab-based techniques, as opposed to the limited synchrotron facilities, makes lab-based X-ray CT a promising avenue for expediting microstructural analysis and enhancing electrode design. This technique enables the examination of large sample volumes at high resolutions, thereby facilitating efficient microstructural analysis and electrode improvement. Laurencin et al. [24] employed hard X-ray nano-tomography for valuable 3D imaging in reconstructing the Ni-YSZ cermet support of SOFC. To ensure a comprehensive 3D reconstruction of the SOFC, it is crucial to use hard X-rays with higher beam energy to prevent excessive absorption within the SOFC due to its coarse microstructure [24]. This innovative approach facilitated the exploration of large volumes at intermediate resolution and offered a more practical alternative to 3D reconstructions based on FIB-SEM technique. The latter is impractical for examining large volumes due to its time-intensive nature and the high resolution provided is unnecessary for this specific application [24]. Ultimately, Heenan et al. [25] delved into an investigation encompassing Ni, Ni-YSZ and Ni-GDC by employing synchrotron X-ray diffraction. This aim was to unveil the impact of ceramic additives on the mechanical properties of the metal throughout thermal cycling.

Although several types of tomography mentioned above produce microstructural images with excellent contrast, but the adjacent phases that are in the same phase often has different intensities. Noise in microstructural images is a common problem that can disguise phase boundaries or appear as boundaries that do not exist. Therefore, manual segmentation process relies on the experience of materials scientist to distinguish between noise and true boundaries. The outcome will be quite varied and subjective if the image segmentation area divides into several regions and process by different material scientists [26]. Apart from different types of tomography mentioned above, image processing is one of the promising methods as depicted in Fig. 1 that produces high quality microstructural images obtained by tomography for microstructural quantification analysis and visualization.

Additionally, the extraction of microstructural properties from microstructural images is a complex undertaking process because it entails a series of intricate image processing steps including image filtering preprocessing, image enhancement, color and texture feature extractions, and image segmentation. Next, some of the factors, like low contrast to discriminate between microstructural phases, image noise, poor segmented regions and edges, may also introduce while performing the image processing steps as mentioned earlier. Therefore, for SOFC microstructural analysis, a good and efficient image processing algorithm to process and quantify the microstructural images are essential for best and accurate results in terms of image quality, microstructural properties such as average diameter, volume fraction, TPB, area interface density and tortuosity factor as these microstructural properties play an important role in power generation performance, heat transfer and chemical reactions.

Lastly, this article comprehensively highlights the merits and drawbacks, detail procedures and other aspects of digital image processing and quantification methodologies used to extract and analyze microstructural properties of SOFC.

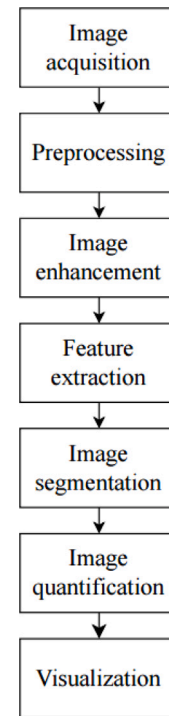


Fig. 1. Flowchart of image processing of SOFC electrodes.

2. Image processing

2.1. Image enhancement

Image enhancement is a method of enhancing the image appearance performance of image processing system. Singh and Mittal [27] discussed on image enhancement using Fourier Transform based techniques as pictured in Fig. 2. The journey of image enhancement using Fourier Transform based techniques commenced with the input image, initiating a sequence that included preprocessing to address issues like noise reduction or resizing. Subsequently, the Fourier Transform was applied to transform the image into frequency domain for revealing spatial frequency components. Next, several types of image filtering such as high-pass filter, low-pass filter or band-pass filter were skillfully employed to enhance the image. The inverse Fourier Transform was applied to reintegrate the modified frequency information back into spatial domain. Furthermore, post-processing steps encompassed contrast adjustment and sharpening are proceeded to improve the image quality. Lastly, the enhanced image demonstrated improvements achieved through the manipulation of frequency components in the Fourier domain.

Table 1 provides a quick application overview of histogram enhancement techniques in several models. The value of thresholding segmentation from the histogram graph based on a certain calculation is referred to histogram enhancement. Next, Agaian et al. [28] discussed three methods included logarithmic transform shifting and logarithmic transform histogram matching and shaping for image contrast improvement. Firstly, the first method mimics the ability to transform the histogram without overexpansion, giving it a considerable speed advantage while the second method utilizes shifting method to map the histogram and return data to spatial domain. However, the image quality will be affected by the loss of image information if the histogram is shifted too far. Lastly, the third method, which is involved in tweaking mean, μ and standard deviation, σ , can be used to adjust image contrast according to needs, but it requires optimal point which increases computational time [28]. In addition, Lee [29] introduced

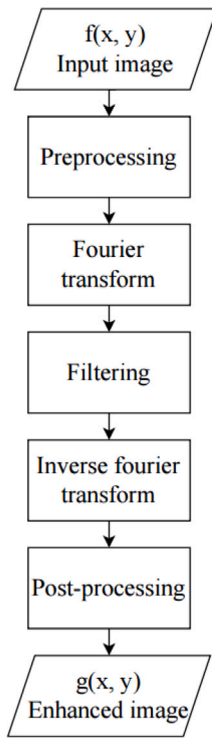


Fig. 2. Flowchart of Fourier Transform based image enhancement [27].

a method for compressing dynamic range of images that focus on retinex theory and discrete cosine transform (DCT). The method uses DCT coefficient to accelerate the processing speed of image enhancement while modifying alternative current coefficients to boost image contrast. As a result, the contrast and image dynamic range have enhanced without over boosting noisy information and introducing block artifacts [29]. Besides that, dynamic histogram equalization that focuses on histogram partition and gray level distribution was presented by Abdullah-Al-Wadud et al. [30]. Due to the allocation of precise, sequential and non-overlapping gray level ranges to sub histograms that will map to the same gray level in output image, the algorithm prohibits under or over enhancement of any region of image and no major loss of image as well [30]. Therefore, the proposed algorithm improves image enhancement effectively without having a negative impact. A new multi-histogram equalization (MHE) framework was discussed by Menotti et al. [31] for brightness preservation and image enhancement. They came to the conclusion that MHE outperforms alternative histogram equalization (HE) methods such as classical HE method (CHE), brightness Bi-HE method (BBHE), dualistic sub image HE method (DSIHE), minimum mean brightness error Bi-HE method (MMBEBHE) and others in terms of image brightness retention and natural appearance yielding after comparing their brightness, contrast and peak-signal-to-noise-ratio (PSNR) value [31].

Besides that, Gonzalez et al. [32] comprehensively discussed on working theory of HE and wrapped up that HE is an adaptive and simple contrast enhancement tool for the images that have significant contrast difference. Besides that, Boudraa and Diop [33] reported a new image enhancement technique through 2D teager-kaiser energy operator (2DTKEO). In contrast to HE, which generates contrast over enhancement and edge-based contrast enhancement (ECE), which produces artifacts owing to noise, teager-kaiser contrast enhancement (TKCE) delivers better image without over enhancement and saturation [33]. Furthermore, histogram modification framework and content adaptive algorithm that was presented by Arici et al. [34] can produce natural looking and enhance image's contrast and video without

decreasing its quality. Nguyen et al. [35] presented an approach for improving video which is constructed by sequence of image frames that involved in reconstructing video from numerous compressed versions of the same content that have varying levels of video quality. The suggested method achieves a decreased image distortion and significant PSNR by utilizing the compressed video information [35]. Besides that, Hu [36] improved the contrast, resolution, noise level reduction, and sharpness of video by applying video enhancement method through content classification and model selection. Dong et al. [37] addressed a reconstructed de-hazing algorithm on inverted low lighting video to produce enhanced low lighting video that preserves critical video information. Lastly, a genetically based technique for enhancing the contrast of low dynamic range image was discussed by Hashemi et al. [38] and they concluded that the respective technique can produce natural looking images and work well in high dynamic range of low illuminated image.

2.2. Color feature extraction

In addition, Table 2 presents the advantages and disadvantages of color feature extraction approaches. Wong et al. [39] introduced merged histogram method (MHM) for obtaining images based on the image dominant color. Although conventional color histogram (CCH) which is invariant to geometrical data like translation and rotation works well in precise color matching through histogram intersection method (HIM), but it has some limitations particularly in red, green and blue (RGB) color space such as low similarity rank for high similarity images and high similarity rank for colors that are completely different [39]. Furthermore, Saad [40] concluded that CCH calculates the probability of image intensities by computationally calculating the number of pixels and indicating color frequency occurrence of image. As a result, it is relatively straightforward and easy to compute, but it has a high degree of dimension without considering the color similarity between bins as well as does not encode any color spatial information [40]. Besides that, Han and Ma [41] discussed on fuzzy color histogram (FCH) for retrieving color image. Unlike CCH, which only assigns each pixel to one of the bins [41]. FCH takes into account the similarity of color by spreading each pixel value across all histogram bins, improves the efficiency in handling dimensions as well as reducing the computational time by implementing FCH clustering algorithm [41]. Based on their comparison, CCH is more susceptible to noise interference than FCH, especially in quantization errors and changes in illumination intensity. In contrast, FCH not only reflects the color similarity belonging to separate bins, but also encompasses on color dissimilarity belonging to the same bins [41].

According to the point of view by Saad [40], FCH uses a fuzzy set membership function to encode the degree of resemblance to other histogram bins. Hence, FCH is more resistant to variation in light intensity and quantization error. Nevertheless, FCH shares the same flaw as CCH that it only distinguishes global image properties and has a high dimensionality [40]. Furthermore, Huang et al. [42] presented color correlogram (CC) for image indexing and retrieval. They emphasized that the advantages of CC, including the ease of computation and small feature size, as well as the fact that CC contains spatial color correlation data that may be utilized to characterize the global distribution of local spatial color correlation [42]. Furthermore, Saad [40] shared the color shape-based method (CSBM) that Howe and Huttenlocher [43] suggested. The K-means technique is used to group image pixels into K clusters and follow by image quantization on original image to obtain quantized color image. Perimeter-intercepted lengths (PILs) is carried on to describe the identical color of connected region by intercepting the perimeter of a connected region with eight lines segments, orientations and passing through the center of region [40]. Therefore, Saad [40] concluded that CSBM proficient in encoding colors and shapes but sensitive to contrast and noise level.

Table 1
Histogram enhancement techniques.

Author	Operating domain	Model	Processing methods	Application
Agaian et al. [28]	Spatial domain	HE based Logarithmic transform histogram shaping using Gaussian distributions (LTHSG)	Logarithmic transform histogram matching and shaping; Logarithmic transform shifting	Security surveillance; Traffic monitoring
Lee [29]	Spatial domain	Dynamic range and contrast compression	Retinex theory; DCT	Image compressing
Abdullah-Al-Wadud et al. [30]	Spatial domain	Dynamic HE on partitioned and allocated images	Histogram partition; Gray level allocation	Image processing
Menotti et al. [31]	Spatial domain	MHE framework	Minimum within-class variance MHE; Minimum middle level squared error MHE	Image processing
Gonzalez et al. [32]	Spatial domain	HE	Global HE	Image processing
Boudraa and Diop [33]	Spatial domain	2DTKEO	Selects the edge strength value of highest energy activity of pixel	Image processing
Arici et al. [34]	Spatial domain	HE based modification	Global contrast enhancement; Histogram modification framework; Content adaptive algorithm	Image processing; video display
Nguyen et al. [35]	Transform domain	Cauchy distribution model	Video reconstruction of AC transform coefficient by using narrow quantification constraint sets	Compressed video
Hu [36]	Spatial domain	Content adaptive video processing model	Content classification; Quantized approximation; Adaptive processing	Image processing
Dong et al. [37]	Spatial domain	Video enhancement algorithm for inverted low lighting video	State-of-art image de-hazing algorithm; Temporal correlations between subsequent video frames	Low lighting video
Hashemi et al. [38]	Spatial domain	Improved contrast enhancement model	Genetic algorithm	High dynamic range of image processing

Table 2
Advantages and disadvantages of color feature approaches.

Color feature	Advantages	Disadvantages
CCH [39,40]	Encodes spatial info; computes the probabilities of color intensities by counting the number of pixels; ease of computation; translation and rotation invariant.	High dimensionality of color space; does not encode color spatial information; does not consider color similarity; low similarity rank for high similarity images; high similarity rank for absolute different color.
FCH [40,41]	Computes pixel color similarity; high resist to quantization errors and contrast changing; improves efficiency in handling dimensions and computation; emphasis on color dissimilarity.	High dimensionality of color space; does not encode color similarity.
CC [40,42]	Encodes local and global spatial information; ease of computation; small feature size.	High dimensionality of color space.
CSBM [40,43]	Clusters pixels using K-means; quantifies area of object using number of pixels; characterizes object shape perimeter through PILs.	Longer computational time; sensitive to noise level and contrast; requires appropriate threshold levels.

2.3. Texture feature extraction

First of all, Patel and Gamit [44] presented extensive analysis on color and texture feature extraction methods for content-based image retrieval. Dilation and rotation are performed on 2D Gabor function by tweaking the Gabor function scale, S and number of orientations, K [40,44]. After that, the image is convoluted using the result obtained from Gabor function [40]. Based on their investigation of various methods for extracting texture feature, including the Tamura feature, Steerable pyramid, wavelet transform and Gabor wavelet transform, they drew the conclusion that the Gabor wavelet transform achieves the best texture retrieval results but is computationally demanding due to multi-orientation and multi-resolution [44]. On the other hand, contourlet transform which is constructed by Laplacian pyramid and directional filter bank (DFB) [40] was introduced by Feng et al. [45] to improve images with poor dynamic range and grey level contrast. In order to compare the performance of enhancement, they conducted

an experiment using various sorts of approaches, including HE, local normalizing, linear unsharp masking, wavelet transform and contourlet transform. Lastly, they summarized that contourlet transform effectively controls the noise level and able to preserve thin and faint contours [45].

Additionally, shift-invariant complex directional filter bank (CDFB) was proposed by Vo et al. [46] for texture image retrieval. The resolution level of CDFB comprises a Laplacian pyramid and a pair of DFBs, known as primal and dual filter banks as well as the corresponding filters in primal CDFB has Hilbert transform correlation [46,47]. In concise, they clearly asserted that CDFB outperforms Gabor wavelet transform and Steerable pyramid in terms of shift-invariance, low-redundant ratio and multi-scale and multi-directional transform with high angular resolution. Contradictory to other two texture feature extraction methods stated previously, where the number of directional sub bands cannot be increased after selecting a sub band filter, CDFB can increase the number of scales by replicating the pyramid [46]. Besides that,

Table 3
Advantages and disadvantages of texture feature extraction approaches.

Texture feature	Advantages	Disadvantages
Steerable pyramid [40,48–50]	Multi-scale and self-inverting; supports any numbers of orientations regardless translations or rotations; reduces the amount of aliasing.	Subbands undecimated; requires longer computational time and larger storage; overcompletes by a factor of 4K/3.
Contourlet transform [40,45]	Decimates lower subbands; able to control noise level and preserve thin and faint contours.	Supports only power of 2 for number of orientations.
Gabor wavelet transform [40,44]	Multi-resolutions; utilizes morphological dilations and convolution to achieve highest retrieval results.	Overcompleted representation of image with intensive computation.
CDFB [40,46,47]	Low redundancy ratio; shift invariance; multi-scale and multi-directional with high angular resolutions; can increase the number of scales by replicating the pyramid; competitive texture retrieval results.	Intensive computation.

Heeger and Bergen [48] presented a pyramid-based method for synthesizing the appearance of a texture sample in an image or solid texture. One of the pyramid-based techniques is Steerable pyramid, which performs image decomposition and separation on images and transforms them into multiple spatial frequency bands and a number of orientation bands. Furthermore, Steerable pyramid is a multi-orientation, multi-scale and self-inverting image decomposition because of rotation and translation invariant implication [49]. Comparing between the number of pixels in pyramid and original image, the number of pixels in pyramid are higher when the low-pass band is subsampled. In light of the fact that pyramid-based texture analysis algorithm treats each sub-band independently, this advantage helps to reduce the amount of aliasing within each sub band [48,50]. Other than that, regardless of the non-orthogonality, the reconstruction of synthesis may be computed effectively owing to the Steerable pyramid self-inverting tendency. Nevertheless, the Steerable pyramid has a distinct downside in that it underperforms in terms of computational efficiency and drastically overcomplete by a factor of 4K/3 [48,50]. Lastly, Table 3 presents the advantages and disadvantages of texture feature extraction approaches.

3. Image segmentation

3.1. Threshold based image segmentation

The adaptive threshold segmentation algorithm (ATSA) was then introduced by Hu et al. [51] to segment asphalt concrete images based on image radius. They observed that the distribution exhibits a gradual increase in gray values from the image center towards the boundary because the non-uniform of asphalt concrete images attributed to ray attenuation, noise and scatters. Due to discernible differences in gray levels among the aggregates and air-voids within a defined annular region, an improved Otsu's thresholding technique proficiently utilized to distinguish the aggregates and air-voids from the background image and obtain microstructural properties by calculating optimal threshold for each annular area of the asphalt concrete images [52,53]. Recently, Zhao et al. [54] explored the efficiency of Otsu's thresholding during pre-processing stage of image binarization. The original image of titanium alloy, $\text{Ti}_6\text{Al}_4\text{V}$ is directly segmented using Otsu's thresholding [55] without any pre-processing steps like filtering, contrast adjustment and others. However, the bright regions of α that diffracted strongly are incorrectly segmented as β [54]. This is due to the fact that the contrast of $\text{Ti}_6\text{Al}_4\text{V}$ image captured by BSE tomography not only includes crystallographic diffraction contrast but also, atomic number variation in the α and β phases as a result of partitioning of V and Al. Over this, it is crucial for automated grayscale thresholding to deliver accurate results because the microstructural images that were captured have uneven grayscales [54]. So, Zhao et al. [54] reformulated an automated pre-processing method for image contrast optimization which includes subsequent steps; applying low-pass Gaussian filter to distinguish objects from the background image, applying 3×3

kernel 2D Median filter for noise suppression and optimizing grayscale contrast plot. Following the implementation of Otsu's thresholding, the α and β phase segmentation accuracy have enhanced [54]. Additionally, Otsu's thresholding exhibits a notable advantage by meticulously computing a criterion that serves as a qualitative measure for the data segmentation [56]. This criterion is derived from the ratio of between-class variance to the overall variance encapsulated within the volumetric data [56].

Furthermore, on the rock properties, the effects of threshold image segmentation and image voxel size were studied by Saxena et al. [57]. Based on image processing and quantification analysis on 2D microstructural images, the porosity values are plausible by using Otsu's thresholding [55] with HSV segmentation algorithm and K-means clustering colored segmentation [58]. However, Otsu's thresholding on red-blue channels exaggerates the porosity level. Furthermore, colored rock images are artificially coarsened using nearest-neighbor interpolation by increasing the voxel size and segmenting accordingly for investigation of pore size and permeability. After that, Saxena et al. [57] calculated the pore size by utilizing the algorithm suggested by Legland et al. [59] which demonstrates the drastically increment effect on size of voxel and concluded that when voxel size increases, pore size also increases significantly. Although watershed transform [60] offers a thorough and excellent image segmentation result that includes every phase of the material, but the identification of regions in proper phase is achieved by thresholding that utilizes the phase separation through intensity difference. As a result, some of the phases might misinterpret due to inconsistent illumination. Lastly, an adaptive variant of Otsu's thresholding [61] approach is applied by Campbell et al. [26] to automate the process of determining suitable threshold for each phase. However, it is imperative to acknowledge that vulnerability of Otsu's thresholding to the challenges posed by noise and varying illumination. The determination of optimal threshold value may prove to be a formidable task if the SOFC microstructural image is noisy and lack of uniformity in illumination. Considering the nature of SOFC which is characterized by its possession of multiple phases, Otsu's thresholding may encounter difficulty when segmenting images with more than two phases.

3.2. Edge based image segmentation

An automated microstructural analysis algorithm for titanium alloy, $\text{Ti}_6\text{Al}_4\text{V}$ was presented by Campbell et al. [62] to overcome inconsistent segmentation result for marker-based watershed transform segmented microstructural images. The optimal threshold values are automatically determined based on image properties by Otsu's thresholding, which maximizes the inter-class variance and is frequently used for image binarization, thresholding and segmentation [32,52,53]. Nonetheless, certain pixels might be misdetected during segmentation due to image noise. Thus, Gaussian filter is used to reduce image noise by smoothing the change of intensity between adjacent pixels [63]. Oversegmentation

is minimized while simultaneously improving the accuracy of edge detection and thresholding with the usage of Gaussian filter. However, the limitation of implementing this filter is the grain boundaries are difficult to retain. According to the proposed method, grain boundaries are marked by edge detection based on intensity differences and edges in small regions on each grain are expanded by morphological dilation. By taking into account factors like grain size, average intensity level between adjacent grains and distance between area boundaries, a split-and-merge strategy is used to merge neighboring regions and resolve oversegmentation error [62]. Comparing the proposed method by Campbell et al. [62] with manual segmentation, the proposed method exhibits superior capabilities in efficiently managing complex structures and intricate details, rendering it well-suited for tasks demanding a comprehensive analysis of the entire image. Conversely, manual segmentation faces challenges in handling fine details or ambiguous boundaries. This method heavily relies on human resources, time and the expertise level of the material scientist, adding to the nuanced considerations inherent in its application.

Next, Cheng et al. [64] presented a multiscale hierarchical segmentation method through joint distribution of intensity and noise-resistant local binary pattern (NRLBP) to identify subtle microstructure tissue sample image region with different textures such as intestine, cartilage and muscle [64]. However, during the analysis, there was a discrepancy between image size and screen resolution and the presented method has analyzed the gigapixel images with different textures by highlighting the regions respectively for visual feedback by users [64].

On the other hand, Nelson et al. [65] published a skeletonization algorithm for image processing and microstructural characterization. Prior to skeletonization, the 3D X-ray of redox stable SOFC electrode, $\text{Sr}_2\text{Fe}_{1.5}\text{Mo}_{0.5}\text{O}_{6-\delta}$ (SFM) images are filtered using Gaussian filter for fuzziness reduction and outliers removal from imperfect alignment during reconstruction process as well as applying sharpening filter to maintain the contrast of the images [65]. To increase the level of smoothing, the filtered images are filtered again using Gaussian filter and then subjected to a mean shift. After that, the images are segmented using a clear and simple grayscale thresholding algorithm. All the pre-processing were undertaken using ImageJ software [66]. In order to avoid the formation of excessive edges at the surrounding of closed pores and excessive unachievable partitioning of solid phase volume during skeletonization process, these pores are filled [65]. Besides that, distance map thinning is implemented for facilitating the imposition of model boundary conditions in Avizo [67] by using algorithm discussed by Sato et al. [68] for skeleton extraction [65]. Therefore, in order to verify that the solid phase will always be represented, the extraction from edges, vertices and segment connectivity are updated [65].

3.3. Region based segmentation

Additionally, Li et al. [69] investigated a novel split-and-merge edge based superpixel segmentation algorithm. The proposed approach has been shown to be successful despite it is a well-known traditional image segmentation method. This is why it differs from another two types of superpixel segmentation algorithms, namely energy optimization and pixels clustering. The geometrically conforming centroidal Voronoi Delaunay triangulation (CFCVDT) algorithm [70], which benefits from fixed runtime regardless of the size of image is used to partition the image into small segments with homogeneous size and shape before performing split-and-merge method [69]. Then, by overlapping the boundaries of each superpixel and using the best splitting contours to maintain the fundamental properties such as color and size, splitting contours are generated for each superpixel using the edge map. For instance, the superpixel should split along the splitting contour as evenly as possible, and the difference between the two regions should be the greatest. Thus, energy function is defined to select the optimal splitting contour based on Bhattacharyya color histogram distance and ratio between areas of two superpixels split by the contour [69]. After

splitting the superpixels, the adjacent superpixels that have a similar small area and color are merged [69].

A novel approach for fluorescence microscopy image cell segmentation, the split-and-merge watershed (SM-Watershed) method was introduced by Gamarra et al. [71] to specifically cater to the nuanced requirements of cell separation. In the initial phase, the marker-controlled watershed algorithm is employed and incorporated diverse preprocessing steps such as erosion, dilation, opening and closing. This phase encompasses pivotal tasks like foreground object marking, background marker computation, and watershed transform computation, resulting in the extraction of cells from the background. Subsequently, the SM-Watershed method as pictured in Fig. 3, operating as a two-stage algorithm, is applied to the segmented image [71]. This method leverages split and merge processes based on the watershed transform, ensuring precise cell separation. Clustered cells are identified by the split process through the analysis of features of the cells, such as area and solidity, followed by the computation of the distance transform before applying watershed transform. Concurrently, in order to identify the oversegmented regions, merge process has employed through the utilization of area and eccentricity. For the cell divisions, morphological operations have applied again to eliminate them [71]. The efficacy of the SM-Watershed method has been substantiated through both quantitative and qualitative visual assessments, confirming its exceptional accuracy and optimal balance between oversegmentation and undersegmentation. Significantly, this method conforms to the irregular shapes of cells, obviating the need for geometric adjustments. When compared to deep learning architectures, the SM-Watershed method outperforms by deep learning architectures. The respective architectures showcased superior performance, demonstrating commendable accuracy against ground truth fluorescence microscopy image especially in original shape of the cells preservation [71]. However, SM-Watershed obviates the need for a training process, making it well-suited for databases with limited data. Additionally, it operates seamlessly without the requirement for labeled data and is applicable to datasets containing only a restricted number of available images. While the SM-Watershed method excelled in identifying cells in fluorescence images, ongoing research is imperative to enhance indicators for missing cells and address any potential noise introduced during the segmentation process [71].

Although the split-and-merge segmentation and SM-Watershed algorithms have not been previously utilized in the analysis of SOFC microstructural images, their effectiveness in handling the intricate and diverse microstructures inherent to SOFC are noteworthy. These effectiveness stem from the utilization of methods especially in overlapping pixel boundaries and the selection of optimal splitting contours, ensuring the essential properties such as color and size are well preserved. In the context of SOFC, which encompasses triple phases (Ni, Pore and YSZ), it is a challenging task to accurately determine the correct phases when dealing with phases overlapping. So, split-and-merge segmentation method emerges as a valuable and strategic solution to overcome this situation. In contrast to manual segmentation, the split-and-merge segmentation approaches demonstrate superior speed in segmentation and computational efficiency. Nevertheless, it is essential to acknowledge their limitations, including the propensity for oversegmentation, undersegmentation, high noise level and only optimal performance with smaller datasets.

To effectively reduce the oversegmentation errors caused by watershed transform [60], Campbell et al. [26] suggested a novel strategy based on region adjacency graph [72]. The concept of region adjacency graph is described in Eq. (1) [26].

$$G = (V, E) \quad (1)$$

where, G = graph; V = vertex of region in image and E = edges.

By employing it, the adjacent regions which belong to the same features that estimated by properties are merged by interconnecting their vertices only when a common boundary is shared as shown in

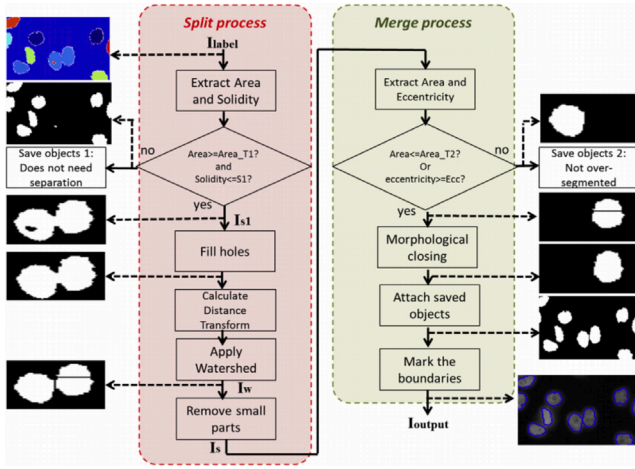


Fig. 3. Flowchart of SM-Watershed algorithm [71].

Fig. 4(a). Instead, the similarity of connected regions plays an important role in weight computation because the edges between similar grain are preserved but internal borders between regions connected by an edge are eliminated when a predefined threshold is exceeded by weight of edges. In Fig. 4(b), the edges between similar grain are preserved but internal borders between regions connected by an edge are eliminated when a predefined threshold is exceeded by weight of edges. Then, the completed area is encapsulated as single grain as pictured in Fig. 4(c) [26]. Since appropriate edge weights and merging thresholds are the crucial criteria to predict the accuracy behavior of region merging, Campbell et al. [26] defined 3 separated weights, w_{1ij} , w_{2ij} and w_{3ij} as shown in Eqs. (2), (3) and (4) [26] as well as 3 individual thresholds, T_B , T_I , and T_S . These declared parameters are used respectively on each pair of regions, so that the results obtained can be compared. In detail, the first weight, w_{1ij} is used to compute the percentage of boundary of smaller grain whereas second weight w_{2ij} targets the intensity difference between each grain. The minimum size of regions can be calculated through third weight w_{3ij} . For thresholds, T_B is used to set as the maximum amount of boundary between grains, T_I assigns as the maximum intensity variation between regions of similar grains and T_S declares as the minimum grain size [26].

$$w_{1ij} = \frac{B_c}{\min(B_i, B_j)} \quad (2)$$

$$w_{2ij} = I_i - I_j \quad (3)$$

$$w_{3ij} = \min(S_i, S_j) \quad (4)$$

where, B_c = common boundary; B = length of boundaries; I = mean intensity of region pixels and S = size of region.

After defining these parameters, merge test, M as presented in Eq. (5) triggered to merge any potential regions that fulfilled the proposed criterion, for example, artifacts and split regions by scratch to an adjacent region [26].

$$M = \begin{cases} 1 & \text{if } w_{1ij} > T_B, \\ 1 & \text{if } w_{2ij} > T_I \text{ or } w_{3ij} < T_S, \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

In the meantime, Joos et al. [73] also demonstrated a region growing segmentation algorithm [74] that was inspired by Jørgensen et al. [75]. The Canny's edge detection is used to determine the phase boundaries prior to image segmentation process and the detected edges are stored in an "edge-map". Then, the starting points, "seeds" for all voxels that most likely belonged to a particular phase is determined.

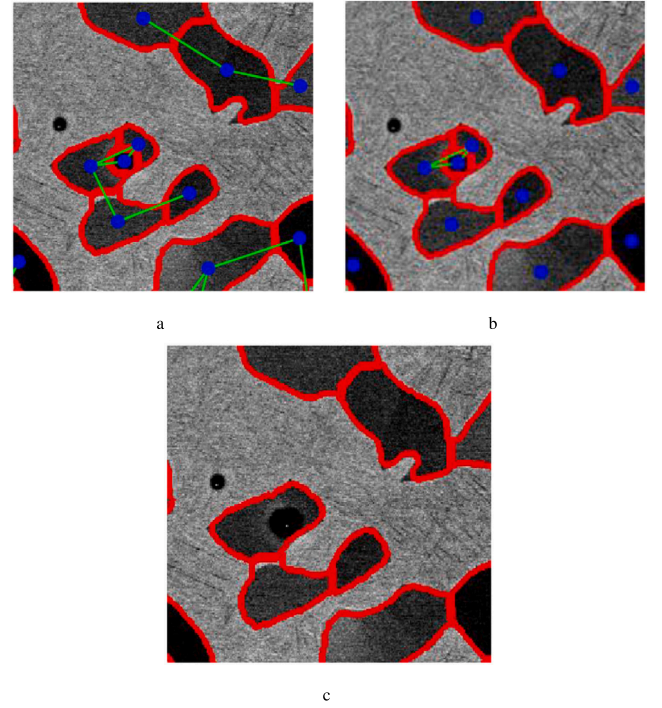


Fig. 4. Illustration of region-based segmentation through region adjacency graph: (a) Schematic histogram at the beginning stage of segmentation, (b) Preservation of edges within similar grains and (c) Encapsulated region [26].

Based on Figs. 5(a) and 5(b), the Otsu's approach [55] for separating pore and YSZ and maximum entropy [76] for separating YSZ and Ni are two traditional segmentation methods that divide the histogram into groups, such as pore seeds, YSZ seeds and Ni seeds for automated seed segmentation. To address the uneven distinctness of gradients across successive images, linear interpolation of determined threshold values is continued [73]. In addition, as the iterative grow process expanded out from the seeds, any of the unassigned nearby voxels that have a joint surface with one of the seed voxels are examined to check whether is it part of the same phase [73]. Therefore, a voxel is added to the seed and subsequent step of iteration is initiated if it is assigned to the same phase [73]. Additionally, morphological operation is applied only after region growing process. As shown in Fig. 5(c), the thin layer of YSZ was wrongly segmented between pore and Ni phase. By applying opening morphological operation, the wrongly segmented layer was removed after erosion and dilation. The downside of region-growing algorithm is robust seeds are required at the early stage of segmentation for the implication of proper edge detection [73].

Bertei et al. [77] released a 3D characterization of SOFC anodes in term of image processing and quantification. Before the deformation, the SEM images that processed using ImageJ [66], Avizo 7.0.1 (Visualization Science Group, Mérégnac, France) [67] and Matlab R2015b (The Mathworks Inc., Natick, MA, USA) [78] were cropped and aligned using least square fit to mitigate the deviation in x and y direction [77]. Then, a $3 \times 3 \times 3$ median filter and an edge-preserving smoothing filter based on non-linear diffusion process were used to improve the grayscale image datasets and threshold-dependent region-growing algorithm with manually assigned seeds was used for image segmentation [77]. After all, volume filtering was used to eliminate voxel-sized artifacts from all phases. On the other hand, a study on image segmentation approach that enables phase boundaries to be identified using geometric criteria rather than gray scale value was presented by Holzer et al. [79]. By using conservative gray scale threshold values, the central region of each phase was discovered and the transition zones pixels were not yet associated with any phases after this stage. Furthermore, the

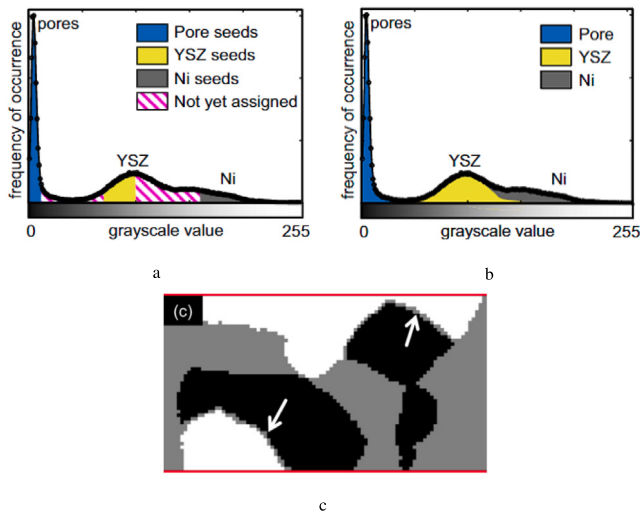


Fig. 5. Region growing algorithm: (a) Schematic histogram at the beginning stage of segmentation, (b) Schematic histogram at the end stage of segmentation and (c) Arrows indicate a thin layer of YSZ generated between pore and Ni phase [73].

asymmetric region growing approach dispersed the transition zones pixels to adjacent phases [79]. The advantage of applying redistribution algorithm is it takes into consideration of the asymmetry of excitation volume at the phase boundary that caused by different atomic numbers of cermet anode elements in SOFC [79].

On the other hand, Chalusiak et al. [80] devised a method for SOFC image segmentation by deploying Particle Swarm Optimization (PSO). First of all, the acquired images are filtered through two distinct stages, namely automated image filtering and post-filtering, aimed at sharpening and smoothening the residual aspects of the images [80]. It is noteworthy that the paper published by Chalusiak et al. [80] includes a thorough explanation of the automated image filtering algorithm, types of filters used, parameters and post-filtering procedures. In addition, Chalusiak et al. [80] ingeniously implemented seeded region growing image segmentation on the filtered images. They also described that the selection of the seed point necessitates meticulous attention to various pertinent criteria, such as proximity to the center of the region, execution of one iteration of image erosion for each phase and others [80].

3.4. Watershed transform segmentation

A technique to segment alpha grains of titanium alloy, $\text{Ti}_6\text{Al}_4\text{V}$ was presented by Campbell et al. [26] utilizing gradient magnitude function as the topographic surface [81]. Before segmenting globular alpha grains, Gaussian filter [81] had applied to reduce microstructural image noise and distortion as well as preserving the intensity difference between adjacent grains from $\text{Ti}_6\text{Al}_4\text{V}$ images [26]. According to datasets analyzed by Campbell et al. [26], SEM $\text{Ti}_6\text{Al}_4\text{V}$ images were filtered using 5×5 kernel Gaussian filter, whereas optical $\text{Ti}_6\text{Al}_4\text{V}$ images or images containing laths were filtered using 3×3 Gaussian filter. Apparently, it appeared that large kernel for filter is needed for SEM $\text{Ti}_6\text{Al}_4\text{V}$ images as opposed to optical $\text{Ti}_6\text{Al}_4\text{V}$ images because SEM $\text{Ti}_6\text{Al}_4\text{V}$ images have higher noise level than optical $\text{Ti}_6\text{Al}_4\text{V}$ images. Even though Gaussian filter is efficient in noise reduction and intensity preservation, it has the drawback of distorting images. So, noise reduction and image distortion must be compromised and adjusted based on requirement to obtain the best outcome [26]. After that, Fig. 6(a), SEM $\text{Ti}_6\text{Al}_4\text{V}$ images are transformed to topographic surfaces as shown in Fig. 6(b). Next, Fig. 6(c) shows the boundaries of alpha grains are positioned where there are the most significant intensity differences between markers. To obtain the areas of the image with

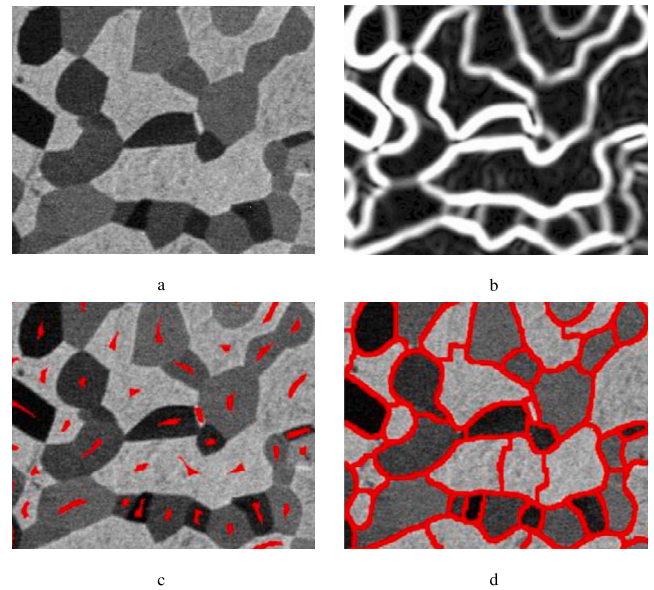


Fig. 6. Watershed transform illustrations: (a) Original image and (b) Topographic surface. Resolved image representations: (c) Markers image and (d) Segmented image [26].

noticeable intensity fluctuations, edge detection was performed [32]. Nevertheless, the magnitude of intensity variation in the image that implies the presence of a grain boundary depends on the edge detection's sensitivity, E_s . In the meantime, constraints on marker-based watershed transform [60] had implemented in order to prevent floods from traveling from one phase of material to another and ensure that each recognized grain only existed in respective phase as illustrated in Fig. 6(d) [26].

Furthermore, Campbell et al. [26] also published a helpful approach for segmenting the colonies of $\text{Ti}_6\text{Al}_4\text{V}$. They discovered that the gradient orientation, computed by Sobel Filter [32] with the greatest change in intensity is measured from positive x -axis in counterclockwise direction. So, determining the gradient orientation of the microstructural image as topographic surface and placing the boundaries closely perpendicular to laths will enable the segmentation in lath of colonies. Upon the placement of markers, globular alpha grain analysis method can be used to segment the lath of colonies and compute the orientation angle of individual laths [26]. Additionally, gradient orientation through pixels was studied and applied to mark colonies. Based on the similarity of orientation values, the pixels are grouped to circumvent laths which are incredibly thin and unclear boundaries. Meanwhile, numerous colonies in the category of bi-modal microstructures were hardly to identify their orientations and frequently visible. Therefore, the laths were divided into smaller regions using the conventional watershed transform [26].

In addition, Ni et al. [82] also discussed image analysis of pore and YSZ for lanthanum ferrite-impregnated SOFC electrode using ImageJ [66,83]. By altering the threshold intensity, the pore layer was cropped and transformed into binary image. Due to the sintering process, a significant number of YSZ grains were in contact, hence watershed transform segmentation is utilized to separate them [83]. Based on the comparison between high magnification image and magnified image with watershed transform segmentation, they came to the conclusion that watershed transform segmented image is closed to high magnification image and the YSZ grains for quantification and spherical objects are well segmented. Watershed transform segmentation also compensated the disadvantages of low magnification images that unable to separate the grain boundary due to low resolutions [82].

Furthermore, Kishimoto et al. [84] published image segmentation algorithm for Ni-YSZ anode of SOFC to evaluate the rate of ammonia

decomposition. First of all, after obtaining the microstructural images by FIB-SEM tomography, the selected regions for microstructural analysis were processed accordingly using Avizo (Thermo Fisher Scientific) [67]. Image smoothing using a combination of Canny edge detector and Gaussian smoothing filter followed by image segmentation through 2D histogram segmentation algorithm based on watershed transform were applied to the microstructural images. Lastly, 3D pore microstructures were reconstructed for microstructural analysis after manually correcting the segmented images [84].

Apart from that, Ullah et al. [85] also published image processing approaches for 3D microstructural reconstruction. The 2D microstructural images were cropped by selecting a proper region of interest (ROI) and rolling ball algorithm [86] was applied through ImageJ software [66] to rectify the uneven illuminated background. Then, brightness thresholding was used to detect the 2D images grain boundaries and followed by morphological opening to eliminate remaining small spots [85]. Instead of manually drawing the thresholded remaining gaps at grain boundaries on each 2D images due to lack of contrast, watershed transform was utilized to automatically fill up the grain boundary gaps and each grain is given a distinctive identity for further analysis. However, the major drawback of watershed transform is over-segmentation that caused incorrect number of regions of grain division by placing more boundaries [85].

All in all, various image processing methods that discussed could be utilized depending on the circumstances, complexity and conditions of SOFC microstructural images. Likewise, a well processed image for further analysis would not be yielded by only opting one image processing method, particularly in image segmentation when it broadly classifies based on the detection of discontinuity and similarity of images [87].

3.5. Machine learning based segmentation

In order to access the intricate microstructure of carbon deposits in the Ni-YSZ electrode, Sciazko et al. [88] presented 3D reconstruction framework, powered by machine learning algorithms. This inventive methodology employs machine learning based segmentation to facilitate the phase segmentation regardless of resin infiltration. Specifically, phase segmentation is achieved by utilizing semantic segmentation patch based convolution neural network (CNN), which has been shown to yield pixel-based accuracy of more than 96% when compared to manual segmentation [88]. Notably, the neural network has been adapted to encompass multiple inputs, including data from lower and energy selective grid (EsB) detectors, thereby significantly enhancing its predictive power. Additionally, the U-net configuration has been ingeniously modified to ensure the preservation of fine details of the deposited carbon structure, while simultaneously extracting the essential global information pertaining to unfilled pores [88].

Besides that, Hwang et al. [11] introduced an innovative approach to analyze the microstructural properties of Ni-YSZ SOFC anode, by leveraging the power of deep learning and semantic segmentation. The method discussed by Hwang et al. [11] comprises three distinct sections, such as pre-processing ground truth image, deep learning implementation and deep learning post-processing. Before moving on to semantic segmentation, the ground truth image undergoes a series of pre-processing steps, for example, cropping, resizing and rotating to ensure that the ground truth image fulfills the requirements of deep learning model and semantic segmentation [11]. The ground truth image is meticulously selected through a rigorous process and subsequently trained using the state-of-art semantic segmentation algorithm, DeepLabV3+ [11]. Additionally, DeeplabV3+ employs a deep learning CNN that incorporates the potent combination of an atrous convolution and atrous spatial pyramid pooling (ASPP) module with augmentation through image-level feature and batch normalization [90]. This unique approach allows for the effective encoding of rich semantic information from ground truth image into dense features with remarkable efficiency [91]. By utilizing the atrous convolution method, DeeplabV3+

applies varying sampling scales to image to produce high quality of semantic segmented image as well as accelerating the convolution process through Xception [91]. This advanced algorithm ensures that the semantically segmented image maintains the same high resolution as the ground truth image [11]. Thus, by implementing this automated deep learning semantic segmentation approach, the artifacts observed in electron microscopy images are effectively eliminated as well as ushering a new era of image quantification with high accuracy and efficiency. Besides that, Zhang et al. [92] addressed on two cutting-edge algorithms, namely CNN and Generative Adversarial Network (GAN) to improve the segmentation result of Ni-YSZ electrode SEM image which is poor resolution and noisy as well as automating the process of segmentation. By incorporating the potential of machine learning algorithms, this study has presented a more reliable and efficient approach of automating image segmentation on SOFC image. Therefore, machine learning based segmentation has provided limitless possibilities for further advancements in microscopy imaging.

Furthermore, Gayon et al. [89] introduced an approach for generating synthetic 3D microstructures utilizing a Deep Convolutional Generative Adversarial Network (DC-GAN) as depicted in Fig. 7. This innovative approach empowers the model to accurately capture the statistical and morphological attributes inherent in real microstructures. These nuanced characteristics are encapsulated in the weight of discriminator and generator networks following a meticulous training process [89]. Within the framework of DC-GAN, the training dynamic revolves around the interplay between a generator and a discriminator. The generator maps a latent vector to real data space while the discriminator assesses the probability of authenticity. The overarching goal is to align the generator's learned density with the distribution of real data [89]. Initially, the discriminator maximizes its loss function through binary cross-entropy, functioning as a classifier. Subsequently, the focus shifts to the generator, which seeks to minimize its loss function, thereby reducing the log probability discerned by the discriminator [89]. In addition, the development of cutting-edge GAN tailored for the creation of 3D porous structures defining Ni-YSZ anodes within SOFC was pioneered by Kishimoto et al. [93]. This innovative GAN model incorporates a conditional vector to control structural parameters. Next, the GAN model's training unfolds in two stages, systematically optimizing the discriminator and generator sequentially for adversarial loss, coupled with additional refinement to minimize volume fraction loss. The optimization employs the Adam optimizer, a batch size of 32, and spans 100 epochs [93]. Notably, the resulting GAN demonstrates an exceptional ability in generating realistic anode microstructures, closely matching the real anodes, even for unseen configurations in training dataset. Being a fully-convolutional architecture, it offers the flexibility to scale output sizes by adjusting input vector dimensions [89]. Nevertheless, the delicate interplay between latent and conditional vector sizes has minimal influence on output quality and diversity. Despite the current limitation in controlling volume fractions, this GAN model paves the way for a digital twin of electrode fabrication and evaluation processes [93].

Furthermore, Dahari et al. [94] proposed a super-resolution based GAN architecture with tri-fold capabilities: enhancing color photograph resolution, transferring styles between images and generating a 3D volume from a single 2D slice. The primary focus of this architecture is on generating mesostructured phase maps. According to their methodology, it excels in various tasks such as super-resolving 3D mesostructured data, incorporating missing phases, altering existing mesostructured data and capturing anisotropy [94]. Validation of the proposed GAN was conducted across multiple datasets, encompassing isotropic triple phase Li-ion battery cathodes, two phases' separators, and low resolution triple phases fuel cell anodes. The result demonstrated remarkable agreement between metrics calculated on 3D high resolution ground truth microstructural images and super-resolved outputs, especially in transport efficiency metrics. This underscores the

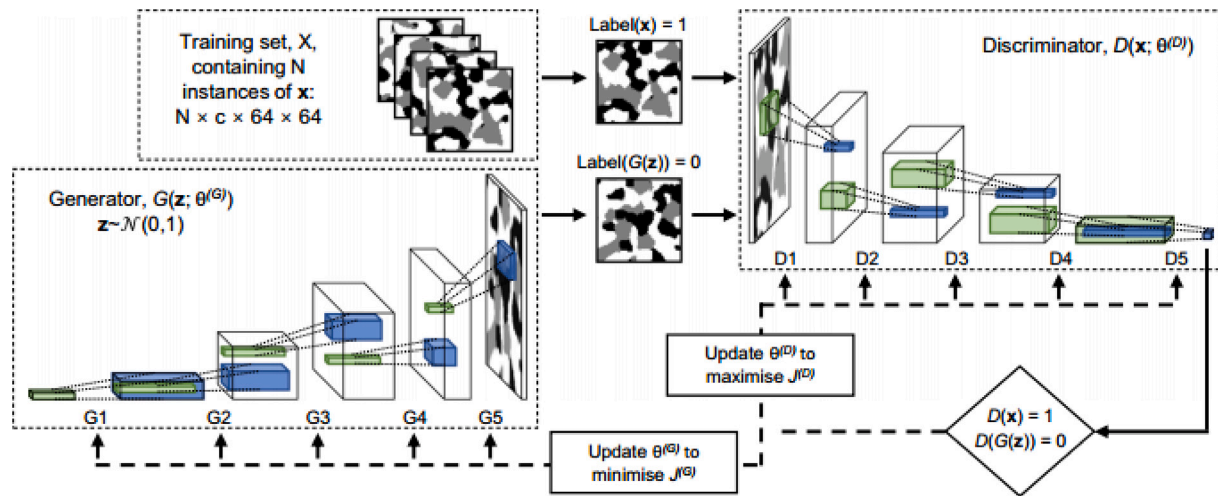


Fig. 7. Architecture of DC-GAN for 2D microstructural images. The green sub-volume indicates a convolutional kernel at a random position within the volume and the blue sub-volume indicates the resulting convolution outcome. The kernel shares the same depth as the one non-spatial dimension but traverses the two spatial dimensions to construct the subsequent layer of the image [89]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

potency of GAN for dimensionality expansion and data fusion, addressing limitations in low resolution 3D and high resolution 2D data [94]. However, it is crucial to acknowledge the inherent training instability of GAN [89]. An example is the discriminator's tendency to overfit on high resolution 2D data, potentially compromising the generator's performance in mesostructured data generation. This highlights the delicate balance required in GAN training to achieve optimal and reliable results [94]. In addition, Zhang et al. [95] explored an inventive approach for 3D reconstruction of LSM infiltrated YSZ electrodes using multi-stage distance correlation function recovering method and a representative nano-resolution 2D micrograph with super-resolution. Instead of duplicating the exact microstructure, this approach prioritizes capturing statistical significance [95]. Additionally, various geometric parameters such as tortuosity factor, area interface density, TPB length and others were systematically computed from the 3D reconstruction. These parameters were then quantitatively correlated with evolving characteristics of the electrode impedance spectrum through distribution of relaxation time analysis across various LSM infiltration loadings [95].

Besides that, Liu et al. [96] implemented a novel deep learning framework that integrates the finite element method and 3D CNN to forecast the macroscopic properties of porous Ni-YSZ anodes. The procedure involves generating 3D microstructures using a combination of discrete element method and Potts kinetic Monte Carlo methods. Subsequently, finite element calculations are employed to determine macroscopic property such as elastic modules, shear modules, Poisson's ratio, coefficient of thermal expansion and TPB density [96]. These microstructures and corresponding macroscopic properties constituted the dataset for training the CNN model, which is subsequently evaluated and applied to predict properties in a testing dataset. Within the CNN framework, pivotal components encompass convolution layers, pooling layers and fully connected layers. Convolution layers extract features from microstructural images using kernels while pooling layers efficiently reduce dimensions of microstructural images. Drawing insights from Liu et al. [96], the study delves into the batch size, a key hyperparameter is explored within intervals from 10 to 60, impacting mean absolute percentage and R^2 score. The kernel size within the convolution layer emerges as another crucial hyperparameter affecting prediction accuracy. Despite encountering lower errors in training datasets, the CNN demonstrates robust generalization [96]. However, it is observed that within the same training epochs, the computational time increases with larger kernel sizes. Lastly, the CNN model excels in both accuracy and time efficiency, taking 38 h for training with 1347 microstructural

sample images and only 14 s for predicting the outputs of 347 microstructural testing images, significantly outshining traditional finite element calculations [96]. Niu et al. [97] introduced a revolutionary performance-informed deep learning framework known as π learning. This framework leverages GAN and CNN in conjunction with advanced multi-physics, multi-scale modeling for 3D porous electrodes [97]. The π learning approach incorporated crucial physical features to facilitate microstructure generation, ultimately predicting overall performance for candidate electrodes. This methodology has been validated for SOFC through two design philosophies such as forward and inverse designs. Notably, the effectiveness of π learning is demonstrated in generating realistic 3D electrode microstructures with targeted performance in inverse design [97]. Furthermore, it successfully identified the globally optimal electrode microstructure, maximizing operating current density at specified overpotentials. The versatility of π learning structure allowed for translating physicochemical insights into rational and cost-effective electrode design, thus advancing cross-paradigm electrode design through physics-informed deep learning across various electrochemical energy applications [97]. The challenge of ionic conductivity in SOFC electrode design is addressed through a comparative analysis involving the optimized electrode microstructure and other samples. π learning proves to be instrumental in identifying two crucial considerations for achieving high performance SOFC electrodes, for example, when adjusting the phase fraction for design, the fractions of pores and YSZ take precedence over the nickel fraction and enhancing the utilization of TPB quantified by the TPB ratio, necessitates minimizing isolated phases [97]. These insights underscore the effectiveness of π learning in addressing key factors for optimizing SOFC electrode performance [97].

4. Image quantification

4.1. Average diameter

Zelew et al. [98] proposed image quantification comparison analysis based on different image processing techniques, such as adaptive enhancement-based thresholding algorithm (AETA) and volumetric-based thresholding algorithm (VTA) embedded by watershed transform segmentation. Both of the techniques are used to calculate aggregate morphological and shape properties, for example, perimeter, area, equivalent diameter (D_{eq}), major and minor axis lengths and others.

Fig. 8 shows the conversion of morphological area to morphological equivalent diameter. The total number of pixels in the aggregate region

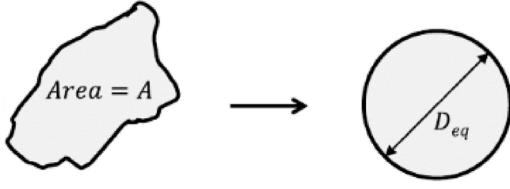


Fig. 8. Conversion of morphological area to morphological equivalent diameter [98].

multiplied by the image resolution was used to determine the aggregate morphological area [98].

The equivalent diameter of an aggregate particle was quantified through Eq. (6). However, for the total equivalent diameters of entire 2D area and 3D volume, Eqs. (7) and (8) were determined respectively by performing summation of equivalent diameters [98,99].

$$D_{eq}(i) = \sqrt{\frac{4A(i)}{\pi}} \quad (6)$$

$$D_{eq}(2D) = \sum_{i=1}^M D_{eq}(i) \quad (7)$$

$$D_{eq}(3D) = \sum_{j=1}^N \sum_{i=1}^M D_{eq}(i) \quad (8)$$

where, $D_{eq}(i)$ = equivalent diameter of aggregate particle i ; $A(i)$ = area of aggregate particle i ; M = total number of aggregates in X-ray computed tomography (CT) slices and N = total number of X-ray CT slices.

Based on the comparison made by Zelelew et al. [98], AETA method has better performance than VTA watershed transform segmentation method in term of separation between connected and overlapped aggregates of the nine asphalt concretes but AETA method cannot preserve the volumetric properties of the microstructures. Campbell et al. [26] discussed quantification method of mean grain size for grains and colonies. After performing phase separation through watershed transform segmentation and other image processing techniques, connected component had configured to measure microstructural properties. In the paper, Campbell et al. [26] described that an ellipse had fit into the region of microstructure to measure the length and width for each microstructure. Average diameter was computed based on the measured parameters by using Eq. (9) [26].

$$GS = \sum_{n=1}^n \frac{L_n + D_n}{2} \quad (9)$$

where, GS = mean grain size; L_n = length of grain n and D_n = width of grain n .

Kishimoto [100] presented a line intercept method as shown in Fig. 9 to quantify the complex microstructure pore size distribution. The microstructure pore diameter was calculated by measuring and averaging the length of the lines intercepted which were drawn onto the pore particle using Eq. (10) [100].

$$d = \frac{1}{3}(r_{x+} + r_{x-} + r_{y+} + r_{y-} + r_{z+} + r_{z-}) \quad (10)$$

where, d = diameter of particle pore size and r = radius from the focused voxel.

Vijay [8] published Ni and YSZ particle sizes quantification using the line intercept method. Following the segmentation and binarization of triple phases of Ni, pore and YSZ respectively, 100 random lines with random angles were drawn from random points along an edge. Four of the image edges were used to generate random lines that span the entire image uniformly. So, the image contains a total of 400 randomly generated lines for particle size computation. As such, the random lines will stop at intersection points where they reached

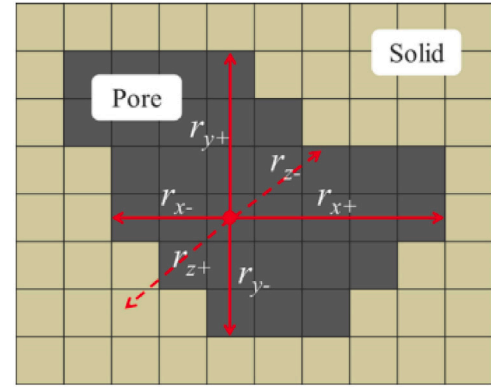


Fig. 9. Line intercept method [100].

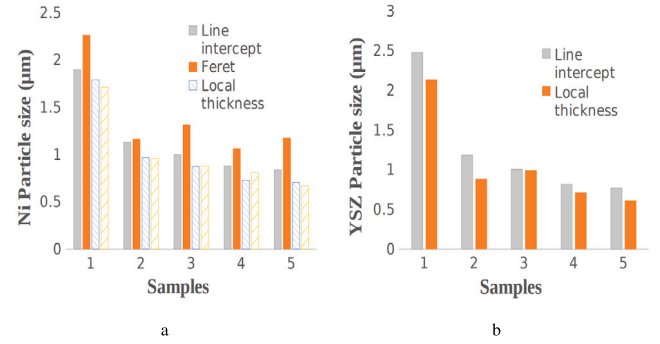


Fig. 10. Comparison of different methods for particle size determination: (a) Comparison of Ni particle size quantification methods and (b) Comparison of YSZ particle size quantification methods [8].

any particular image edges. Hence, the lengths of random lines were spontaneous as well. Lastly, particle size quantification was performed through Eq. (11) [8,18,101].

$$D_i = 1.5 \frac{LV_i}{N_i} \quad (11)$$

where, D_i = particle size of phase i ; L = total length of random lines; V_i = volume fraction of phase i ; N_i = number of grains of phase i crossed by random lines and factor 1.5 = equivalent spherical diameter of the grain.

Additionally, validation of the particle size of Ni and YSZ results using line intercept method in contrast to other methods such as Feret diameter, equivalent diameter and ImageJ [66] local thickness are discussed and illustrated in Figs. 10(a) and 10(b) [8]. Feret diameter which is measured with a caliper is the separation between tangents to the particle profile or distance between two parallel lines drawn on opposite ends of particle at any given direction. For equivalent diameter, it is calculated by measuring the diameter of a circle that corresponds to the particle area as shown in Eq. (6). The advantages of equivalent diameter are simple and not affected by particle shape. When local thickness is utilized in ImageJ [66], it is calculated as the diameter of the largest circle that contains a particular point and fills it entirely. Based on Vijay's [8] findings on particle size of Ni result, he discovered that the line intercept method provides an average result when compared to the results assessed by other approaches as mentioned earlier. However, the line intercept method produces substantially larger particle sizes than the local thickness method. Due to the continuous morphology of YSZ, Feret diameter and equivalent diameter were not applied for YSZ particle size quantification as the particle size cannot accurately reflect [8,99].

Bae et al. [102] also discussed on image processing and quantitative analysis of microstructure of SOFC Gadolinium-doped ceria

GDC/LSM cathode. Next, by computing mean line intercept on 2D thresholded images, particle size of each phase were determined. On the 2D images, at least 91 random lines were appropriately aligned in the horizontal, perpendicular and diagonal directions. Following the mutual connection of the random lines with the relevant phases, mean line intercept was only quantified if the random lines were located within the relevant phases [102]. In addition, Bae et al. [103] also published research about the microstructural characterization of SOFC composite electrode materials using image processing. After obtaining the binarized BSE images which included YSZ-Ni anode functional layer, YSZ-Ni anode layer and GDC layer, line intercept method was utilized again by drawing 91 lines which were distributed in 24 lines in horizontal, 29 lines in perpendicular and 38 lines in diagonal directions onto images [103]. After all, the findings of line intercepts were then used to determine mean line intercepts. Furthermore, a morphological operation (MO) based method [104,105] as described in Eqs. (12) and (13) for determining Ni-YSZ anode particle size that are based on mean size and particle size distribution were presented by Faes et al. [101]. By gradually increasing the opening order, it is feasible to identify the particle size distribution of a complicated microstructure. Knowing the individual surface which is defined by step number and structuring element, a number of particles on surface are lost after each step changes. Faes et al. [101] also emphasized that the class interval step is $0.175 \mu\text{m}$ with an octagon serving as structuring element and at a magnification of 2000.

$$l_{MO} = \frac{\sum_{j=1}^n N_j l_j}{N_{tot}} \quad (12)$$

$$\delta(l_{MO}) = \sqrt{\frac{\sum_{j=1}^n N_j (l_j - l_{MO})^2}{N_{tot} - 1}} \quad (13)$$

where, l_{MO} = mean size; N_j = number of particles analyzed; l_j = particle size of class j and N_{tot} = total number of particles analyzed.

In addition, Holzer et al. [79] discussed on characterization of microstructural degradation of cermet anodes for SOFC through phase size distribution (PSD). They proposed three different geometrical conceptions for PSD, including discrete PSD_D, intercept technique and continuous PSD_C. The geometrical approach for PSD_D as illustrated in Fig. 11(a) was based on the perception of different objects and measurements were made for each of these 2D area or 3D volume. Nevertheless, it was difficult to identify phases separately because connected phases of Ni, CGO, and Pore are usually happened in SOFC anodes [79]. For PSD_D 2D quantification, stereological correction must apply but uncertainties might introduce due to poorly defined stereological correction whereas 3D data obtained from high resolution FIB tomography was required for PSD_D 3D quantification. Next, line intercept method that utilizes randomly oriented intercepted lines was used to measure phase size. Based on Fig. 11(b), even when a bundle of intercepted lines of various lengths and random orientations were drawn on complicated particle shape for PSD, the resultant large size distribution may be challenged to interpret the relationship between microstructural properties [79]. Next, continuous PSD_C as shown in Fig. 11(c) was introduced by Holzer et al. [79], which can be used to characterize continually percolating phases. A 2D circle or 3D sphere with a predefined radius, r_{xyz} was used for measuring the 2D area or 3D volume of each phases. By increasing the radius of 2D circle or 3D sphere, it can fill up a larger 2D area or 3D volume. On the contrary, some narrow corners and edges can be intruded by decreasing the radius. Continuous PSD_C was selected by Holzer et al. [79] because it had outperformed compared to other approaches that mentioned earlier. Based on their analysis, continuous PSD_C does not restrict on object recognition and clearly represent size measurement at each point of image, so, it can apply on continuous phases. The result obtained from continuous PSD_C from 2D images are comparable to the outcomes obtained from 3D tomographic data without the needs of any stereological correction [79].

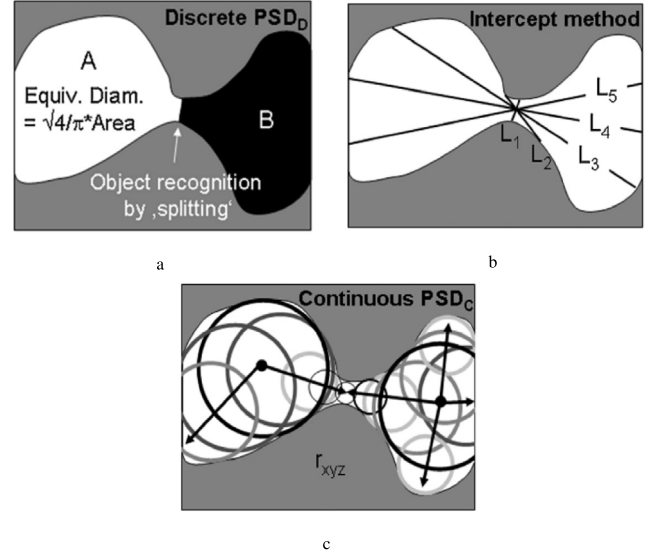


Fig. 11. Geometrical properties of phase size distribution: (a) Discrete PSD_D, (b) Intercept method and (c) Continuous PSD_C [79].

4.2. Volume fraction

An automated image processing method to quantify volume fraction of microstructure after phase separation was proposed by Campbell et al. [26]. The same procedure used to determine average diameter of microstructure was utilized to determine volume fraction when measuring microstructure characteristics. After each grain length and width had been measured, the globular volume fraction was determined through Eq. (14) [26].

$$VF_G = \frac{\sum_{n=1}^n \begin{cases} 1 & \text{if } \frac{L_n}{D_n} > 2, \\ 0 & \text{otherwise.} \end{cases}}{n} \quad (14)$$

where, VF_G = globular volume fraction; L_n = length of grain n and D_n = width of grain n .

Next, Kishimoto [100] also explored manual quantification method to calculate volume fraction of microstructure which has strong relationship with other microstructural properties using Eq. (15). The number of voxels relevant to each phase were counted and divided by total number of voxels in the whole region [100].

$$V_l = \frac{Vol_l}{Vol_{all}} \quad (15)$$

where, V_l = volume fraction of phase l and Vol = volume.

Vijay [8] and Rathore [99] also proposed stereological volume fraction quantification method through Delesse-Rosiwal law to measure volume fraction of phase that was derived from image. Based on Eq. (16), it is clearly described that the volume fraction of a phase is the ratio of phase in term of number of pixels to the total number of pixels in the image [8]. In addition, cumulative length of lines or number of points intersected with the phase of microstructural image can be measured using superimposed line grid or point grid for volume fraction quantification [8,99,101].

$$V_v = A_A = L_L = P_P \quad (16)$$

where, V_v = volume fraction of phase's interest in 3D volume; A_A = area fraction of phase's interest divided by total area of image; L_L = length of lines intersects with phase's interest divided by total length of lines on image and point fraction intersects with phase's interest divided by total number of points on image.

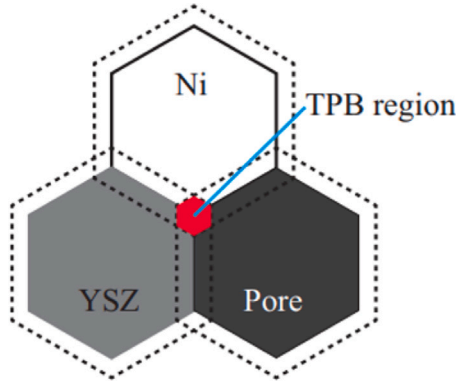


Fig. 12. Illustration of TPB [100].

Raka [106] discussed a mathematical interpretation as described in Eq. (17) to calculate volume fraction of each phase of microstructure. In order to obtain area fraction of each segmented image in the stack of images, the number of pixels of particular phase were counted and multiplied by the area of each pixel. After that, the sum of all the areas were then multiplied by the distance between two images [106].

$$\phi_{phase} = N_{pixel} * X_{pixel} * Y_{pixel} * Z_{pixel} \quad (17)$$

where, ϕ_{phase} = volume fraction of phase; N_{pixel} = number of pixels and $X_{pixel}, Y_{pixel}, Z_{pixel}$ = pixel dimensions.

Bae et al. [102] also mentioned that surface area of phases can be inferred to be equivalent to 3D volume fraction due to the isotropic nature of constituent phases. At the same time, fraction of contact area can be estimated through experimental results from volume fraction and mean line intercept by using Eq. (18) [102,103].

$$\frac{Vol_i}{S_V^i} = \frac{l_i}{4} \quad (18)$$

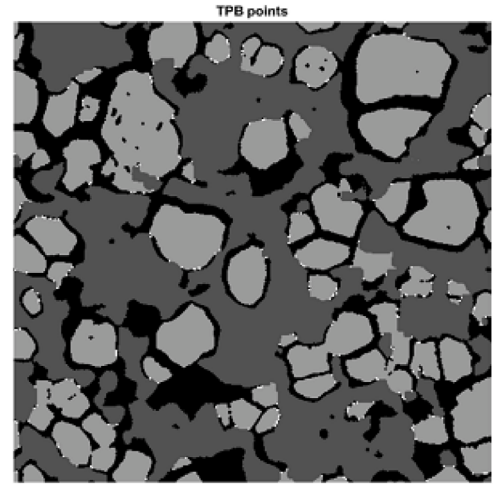
where, Vol_i = 3D fraction of phase i ; S_V^i = fraction of contact area, $\frac{Vol_i}{S_V^i}$ = volume-to-surface-area ratio of phase i and l_i = mean line intercept of phase i .

4.3. Triple phase boundary

On the other hand, Kishimoto [100] discussed a manual centroid counting method to calculate TPB density as illustrated in Fig. 12. After the triple phases of Ni, pore and YSZ were segmented, these segmented areas were expanded outward slightly to obtain overlapping regions among each other. The overlapped regions that contained TPB lines and string-like volumes were extracted. Next, the total length of the TPB lines which reconstructed from the center lines obtained from the string-like volumes were calculated. In order to obtain TPB density, the calculated total length of TPB lines were divided by sample volume. However, if the expansion of volume is infinitely small and spatial resolution of images is sufficiently high, the TPB lines obtained from current approach are theoretically matched with the actual TPB lines [100,107].

In evaluating cell performance and sustaining electrochemical reaction, TPB is crucial. So, Vijay [8] demonstrated the quantification approach for TPB length per unit volume using Eq. (19) [8,99,101]. First, a 2×2 pixel square structuring element was used to scan the microstructural image in order to identify TPB points. Then, the TPB points were determined and counted when the 2×2 pixel square structuring elements contain all three phases as depicted in Fig. 13 [8].

$$\lambda_{TPB} = 2P_A \quad (19)$$

Fig. 13. Identified 2×2 pixel TPB points in white [8].

where, λ_{TPB} = TPB length per unit volume and P_A = points of interest per unit area of image.

The volume expansion method and centroid method were proposed by Iwai et al. [107] for computing TPB density. As illustrated in Fig. 14, two spheres that each represented a phase were placed around the third phase to symbolize the volume expansion working mechanism. Then, a TPB with a circled shape resulted from the partial overlapped of two spheres. On the other hand, they also conducted numerous test calculations by varying the relative positions of two spheres and came up with the conclusion that volume expansion method provides reasonable results for microstructures that have a characteristic length which is greater than 10 voxels [107]. The centroid method approach comprises a series of systematic steps in the computation of TPB density. First of all, the line segment that bordered by four voxels was classified as TPB if the adjacent four voxels consist of every three phases that are initially assigned as Ni, YSZ and pore, and the phases of diagonal voxels are different [107]. Then, the adjacent three midpoints of TPB segments were used to define triangles. Lastly, the distance between triangle centroids were used to compute TPB length. Based on the sample used in the study, the estimated TPB density by volume expansion method is $2.487 \mu\text{m} \mu\text{m}^{-3}$ whereas the estimated TPB density by centroid method is $2.556 \mu\text{m} \mu\text{m}^{-3}$. After comparing the results of both approaches, the result difference is less than 3% [107]. Other than that, Holzer et al. [108] mentioned that total TPB can be measured either from 2D images or 3D tomography data, but it cannot differentiate between active TPB or inactive TPB from 2D images. Hence, the status of TPB can only be retrieved from 3D quantification that provides the resources of connectivity of each phases [108].

A TPB points counting principle using Image-Pro [109] software for SOFC anode TPB density quantification was introduced by Khan et al. [110]. In order to calculate the number of TPB points of processed BSE 2D images where the three phases of SOFC are met, they utilized a 7×7 pixels color cube. Then, Eq. (20) was used to calculate TPB density, $L_{V(TPB)}$ in μm^{-2} which is yielded by total TPB points in 2D images and also fulfilled the mathematical stereology that described in Eq. (19) [110].

$$L_{V(TPB)} = \frac{\text{No. of TPB points in 2D image}}{\text{Area of image}} \times 2 \quad (20)$$

4.4. Area interface density

In chemical reactions and heat transmission of SOFC, area interface density is crucial because the larger the area interface density, the higher the heat exchanges and chemical reaction will be taken [7]. In

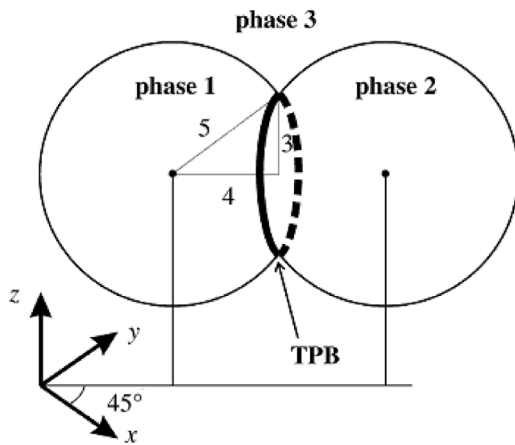


Fig. 14. Geometry set for validation of TPB length through volume expansion method [107].

other study, Kishimoto [100] developed a computational area interface density approach. Based on his study, the boundary surfaces of Ni and pore were measured and subtracted to obtain the intersection area between Ni and pore, which were then used to compute the area interface density of these two materials. After obtaining the area boundary surfaces, they will be divided by phase volume to obtain the area interface density using Eq. (21) [100].

$$(S/V)_l = \frac{S_l}{Vol_l} \quad (21)$$

where, S_l = surface of phase l and Vol_l = volume of phase l .

Vijay [8] and Rathore [99] published area interface density quantification method for Ni, Pore and YSZ individual phases based on stereological relation as described in Eq. (22) [8,99].

$$S_i = \frac{4}{\pi} L_i = 2P_L \quad (22)$$

where, S_i = surface of phase i ; L_i = perimeter of phase i per unit area and P_L = number of intersection points with curves or feature lines per unit length of probe line.

In addition, the surface area of a phase which comprises of areas where it interfaces with other phases were taken into account as shown in Eqs. (23), (24), and (25) [8].

$$S_{Ni} = S_{Ni,YSZ} + S_{Ni,pore} \quad (23)$$

$$S_{YSZ} = S_{Ni,YSZ} + S_{YSZ,pore} \quad (24)$$

$$S_{pore} = S_{Ni,pore} + S_{YSZ,pore} \quad (25)$$

Therefore, due to the fact that a phase surface area can be conceived as the sum of its interface areas with other phases, the area interface density of Ni and YSZ, Ni and pore, and YSZ and pore can all be computed through Eqs. (26), (27), and (28) respectively [8].

$$S_{Ni,YSZ} = \frac{S_{Ni} + S_{YSZ} - S_{pore}}{2} \quad (26)$$

$$S_{Ni,pore} = \frac{S_{Ni} - S_{YSZ} + S_{pore}}{2} \quad (27)$$

$$S_{YSZ,pore} = \frac{S_{YSZ} + S_{pore} - S_{Ni}}{2} \quad (28)$$

Marching cube method is a common approach to quantify area interface density and often used to improve quantification accuracy as well. Matsuzaki et al. [111] employed the marching cube method to quantify the area interface density of voxel structures by utilizing 21 distinct patterns as shown in Fig. 15 to ascertain area interface

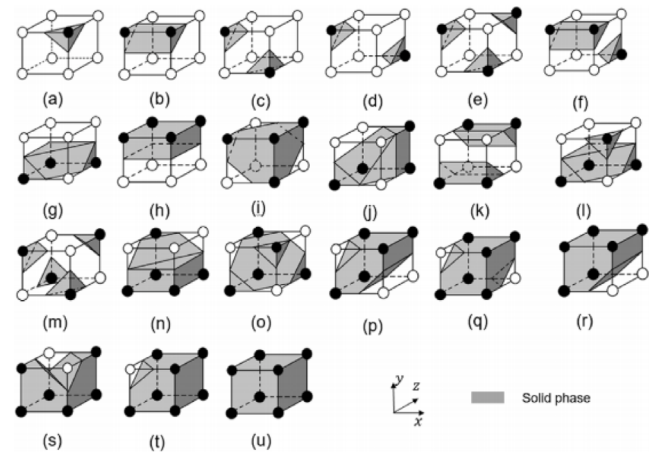


Fig. 15. 21 area interfaces used in the marching cube method for area interface density calculation. Black and white spheres indicate phase information of each voxel and grey indicates the estimated solid phase [112].

density based on the phase information derived from the neighboring 8 voxels. Besides that, the incorporation of marching cube reconstructed structures into Lattice Boltzmann method (LBM) numerical simulation is anticipated to enhance computational accuracy [112]. However, it has come to light that the marching cube method tends to overstate the area interface density by approximately 8%, particularly when applied to spherical structures. In a meticulous comparative analysis, Matsuzaki et al. [111] evaluated the area interface density of the reconstructed microstructure, determined through the marching cube method and compared it with stereology. Next, microstructural samples were categorized along the x -axis, y -axis, and z -axis. The findings revealed negligible variations in area interface densities between x -axis and y -axis divided microstructural samples, while a substantial discrepancy was observed in x -axis divided microstructural samples. This distinction is attributed to potential errors introduced during the image alignment processing in the FIB milling z -direction. Despite the marching cube method demonstrating a commendable alignment with stereology, it showcases a discernible dependence on resolution [111]. To preserve the intricate details of 3D geometries reconstructed through FIB-SEM, a meticulous approach involves defining local effective diffusivity and conductivity was proposed by He et al. [112]. These parameters are established for grids comprising both solid and phase information of each voxel by utilizing the marching cube method as the fundamental technique. Furthermore, in the analysis conducted by Brenne et al. [113], the determination of area interface density was conducted through blurred interface mixture model (BIMM) and marching cube method with subsequent comparison of the obtained results. The area interface density as assessed by BIMM demonstrated a close correspondence to result obtained through marching cube method, particularly in binary ground truth tomography datasets [113]. It is noteworthy that while the marching cube method offers an indication of the actual area interface density within the data, it may not serve as an absolute ground truth due to inherent errors associated with vertex sampling frequency and sample accuracy [113].

4.5. Tortuosity factor

Tortuosity factor signifies the impact of intricate and convoluted pathways of pores that poses a hindrance to the diffusion of Li-ion along a particular direction [114]. The efficiency of transport rate is significantly influenced by the tortuosity factor and volume fraction, notably in porous material. Due to the reduction in conductive volume and intricacy of structural composition of the phase, the transport rates are often lower than in free space [7]. Next, Kishimoto et al. [7]

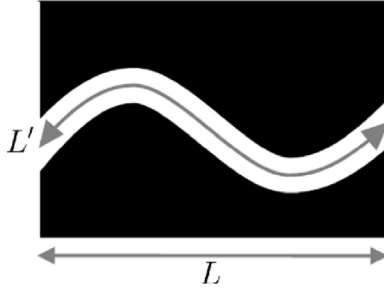


Fig. 16. Curved channel for explanation of tortuosity and tortuosity factor [7].

outlined the distinction between tortuosity and tortuosity factor in Fig. 16. The ratio of length of the curved channel, L' to the length of straight channel, L is known as tortuosity [7], a unitless microstructural properties that could not be averaged using regularly spaced FIB-SEM tomography images [115].

In SOFC, tortuosity is defined as the ratio of the actual diffusion path length and the hypothetical straight channel path. Remarkably, in the x_3 direction, both pores and YSZ microstructures exhibit a tortuosity of 1, indicating a straight and direct path. This stands notably lower when compared to the typical cathode. In cathode, mass and ion transfer primarily traverse the x_3 direction, the low tortuosity signifies a heightened efficiency with minimal resistance facilitating the smooth progression of both mass and ion transfer [116]. The electrochemical reaction within Ni-YSZ is exclusively confined to the anode functional layer while the Ni-YSZ support plays a vital role in ensuring cell's mechanical stability [117]. The percolation of nickel serves pivotal determinant in delineating the electrochemically active area within the cell. It is imperative to demonstrate high electrical conductivity for efficient electron transport. Additionally, it also possess high porosity and low tortuosity to facilitate seamless gas diffusion [117]. Besides that, Wang et al. [118] elucidated that the utilization of 3D reconstruction for tortuosity measurement in the Ni-YSZ anode serves as a valuable approach to assess the prevailing diffusion mechanism, encompassing options like Knudsen diffusion or molecular diffusion. This evaluation is grounded in the effective binary diffusivity obtained through the polarization model [118]. Additionally, tortuosity is quantified by dividing the straight line distance traveled by the estimated length of a gas particle's route as it leaves an intruding gas flow and travels to solid electrolyte [115]. A major tortuosity is a passage that a specific gas molecule must travel through to reach TPB from the gas stream [115]. Contrarily, the tortuosity factor is equal to the square of the tortuosity and describes how the flux of diffusion has increased despite a reduction in effective diffusive cross-sectional area because of path inclination as well as actual diffusion path lengths have increased in comparison to superficial thickness in the mean diffusion direction [7,119]. Nevertheless, it is impossible to precisely define a transport path length due to numerous branches and communications of transport path, therefore, a reliable tortuosity factor is important [7]. Besides that, Kishimoto et al. [120] introduced random walk calculation to determine the tortuosity factor of the SOFC anode. First of all, the pore voxels are randomly assigned by a number of random walkers. Each walker selects one of the neighboring voxels at random as its possible position in the subsequent time step. The walker moves to a selected neighboring voxel that depicts the pore element [120]. On the other hand, the walker remains in the current voxels and waits for the subsequent time step if the chosen voxel is a solid voxel. Not only that, both absorption and desorption are not taken into account during the process and the mean square displacement of random walkers is quantified using Eq. (29) during repetition of the process [120].

$$\langle r^2(t) \rangle = \frac{1}{n} \sum_{i=1}^n \{ \{x_i(t) - x_i(0)\}^2 + \{y_i(t) - y_i(0)\}^2 + \{z_i(t) - z_i(0)\}^2 \} \quad (29)$$

where, $\langle r^2 \rangle$ = mean square displacement; $\langle \rangle$ = ensemble average; $x_i(t)$, $y_i(t)$ and $z_i(t)$ = 3D coordinates of walker's position at time, t for i th walker and n = total number of random walkers.

In addition, the diffusion coefficient, D and time derivative of nonabsorbing particles are correlated due to the $\langle r^2 \rangle$ is proportional to t as shown in Eq. (30) [120].

$$D = \frac{1}{6} \frac{d \langle r^2(t) \rangle}{dt} \quad (30)$$

However, the $\langle r^2 \rangle$ can be differentiated into directional $\langle r^2 \rangle$, for example, $\langle x^2 \rangle$, $\langle y^2 \rangle$, and $\langle z^2 \rangle$ as shown in Eqs. (31), (32) and (33) yet corresponding tortuosities are calculated respectively when porous medium has an anisotropic structure [120].

$$\langle x^2(t) \rangle = \frac{1}{n} \sum_{i=1}^n \{x_i(t) - x_i(0)\}^2 \quad (31)$$

$$\langle y^2(t) \rangle = \frac{1}{n} \sum_{i=1}^n \{y_i(t) - y_i(0)\}^2 \quad (32)$$

$$\langle z^2(t) \rangle = \frac{1}{n} \sum_{i=1}^n \{z_i(t) - z_i(0)\}^2 \quad (33)$$

Besides that, Iwai et al. [107] proposed an Eq. (34) [107] to quantify tortuosity factor when isolated pores are present. Isolated pores which do not contribute to gas diffusion and may prompt overestimation of tortuosity factor are not included in calculation since they resulted in effective pores that are connected to computational domain surface boundaries of stacked images [107]. Additionally, the diffusion coefficient calculation excludes data from the preliminary 5 000 000 time steps because walkers have not yet encountered the porous solid structure at the initial phase of random walk, and 10 000 000 time steps and 100 000 random walkers are chosen in order to enhance the accuracy of tortuosity factor [107]. Lastly, a mirror symmetric border condition is employed to prevent random walkers eluding from the stacked images as the time step proceeds [107]. On the other hand, FIB-SEM reconstructed Ni-YSZ microstructures were used by Kanno et al. [119] to evaluate SOFC anode polarization simulation through 3D LBM. Undeniably that LBM is used to investigate the behavior of a group of particles. Each lattice point in the LBM has velocity linkages with its neighborhood [121]. Lattice arrangement is typically represented by $DmQn$, where m is dimension of issue to be solved and n is the number of connections with other lattice nodes [121]. As demonstrated in Fig. 17, D2Q9 and D3Q19 are two commonly used arrangements. Furthermore, depending on the total number of connections between nodes, a weighting factor is proportionally assigned to each velocity linkages as well as the total sum of weighting factors must always unity [121]. Tortuosity factor, one of the principal SOFC parameters is quantified using LBM. Kanno et al. [119] highlighted that pore and YSZ are essentially isotropic and homogeneous because the tortuosity factor of pore is relatively constant regardless of cross-sectional area whereas tortuosity factor of YSZ behaves the same as pore when cross-sectional area surpasses $200 \mu\text{m}^2$ [119]. Finally, tortuosity factor of Ni exhibits extremely wide fluctuations even in the samples with large cross-sectional area [119].

$$\tau_{\text{Pore}} = \frac{D}{D_{\text{eff}}/V_{\text{Pore}}} = \frac{V_{\text{Pore}}}{V_{\text{Pore,eff}}} \frac{d \langle r^2(t) \rangle}{dt} / \frac{d \langle r^2(t) \rangle_{\text{Pore}}}{dt} \quad (34)$$

where, τ_{Pore} = tortuosity factor of pore; D_{eff} = effective diffusion coefficient; V_{Pore} = volume fraction of pore and $V_{\text{Pore,eff}}$ = volume fraction of effective pore.

It is anticipated that different microstructural properties will be crucial in determining the relationship between electrode microstructure and electrode performance, notably in electrochemical kinetics and catalytic activity in composite electrodes [108]. Therefore, image quantification analysis has a strong and significant association with image processing methods to measure the microstructural properties and obtain accurate result for evaluating the performance and behavior of SOFC.

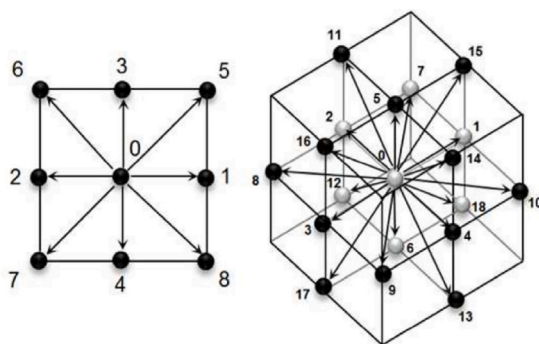


Fig. 17. Most common LBM lattice arrangements. D2Q9 (left) and D3Q19 (right) [121].

5. Conclusion

Image processing has been frequently used to optimize and improve performance of SOFC because it is effective in extracting microstructural properties of SOFC through image quantification. In the perspective of image processing and quantification, this paper not only discussed manual image quantification methods, but also meticulously delivered automated image processing and quantification methods in terms of enhancing identification methods to obtain ideal microstructural properties such as average diameter, volume fraction, TPB, area interface density and tortuosity factor, develop superior algorithms, and boost existing image processing and quantification methods. Last but not least, this study described various types of investigations and experiments that had been carried out by researchers who had went through rigorous analytic methods in order to improve the performance of SOFC through image processing. The future of image processing in SOFC development is promising with numerous opportunities for further advancements. With the ongoing evolution and refinement of image processing techniques, they will play a pivotal role in optimizing SOFC performance, reliability and cost-effectiveness. A particularly noteworthy avenue for exploration lies in multimodal SOFC image analysis, especially the integration of data from various imaging modalities, such as SEM and X-ray tomography, holds substantial potential for offering a comprehensive insight into the microstructure and behavior of SOFC. By combining data derived from multiple imaging modalities, the limitations inherent in each individual tomographic methods are effectively overcome. This integrative approach not only mitigates the limitations associated with tomography methods, but also enhances the overall depth and accuracy of SOFC imaging analysis. Furthermore, the presented algorithms demonstrate remarkable efficacy in tackling the defined research goals and delivering valuable insights to SOFC. However, the implementations of image processing on SOFC are not without limitations as the challenges related to image data quality, computational resources and algorithmic constraints.

CRedit authorship contribution statement

Kar Shen Tan: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Chee Kiang Lam:** Writing – review & editing, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Wee Choon Tan:** Writing – review & editing, Supervision, Methodology, Formal analysis, Data curation. **Heap Sheng Ooi:** Supervision. **Zi Hao Lim:** Writing – review & editing, Methodology, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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